

# Introduction to AI and Data Science

Attendance



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# Introduction to AI and Data Science

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## What is Data Science:

Data Science is the field that uses statistical and computational techniques to analyze data and derive insights or predictions.

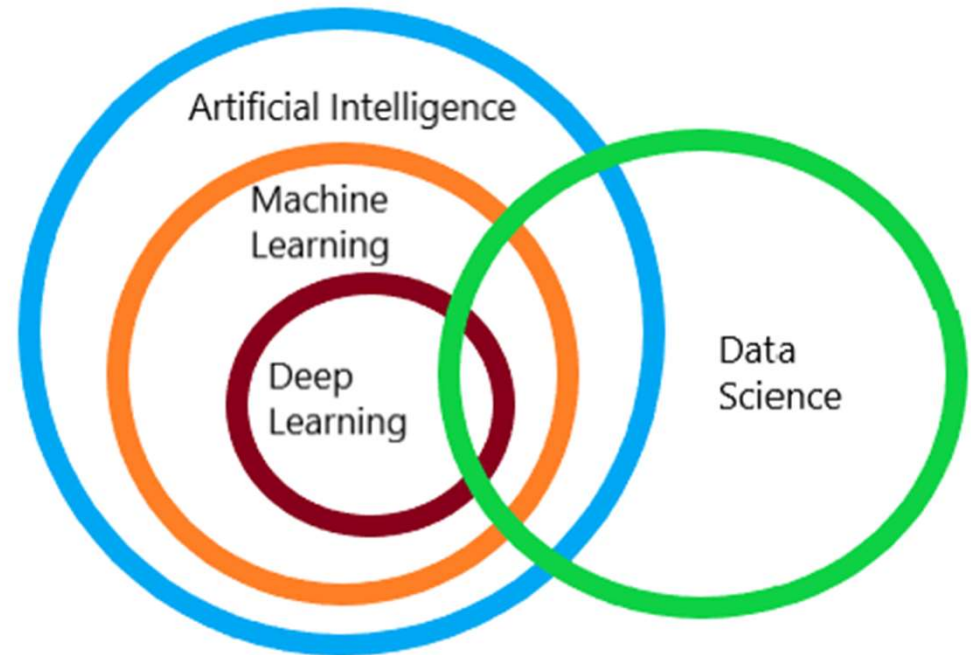
## What is Artificial Intelligence:

Artificial Intelligence (AI) is the simulation of human intelligence by machines.

- enabling them to perform tasks like learning, reasoning, and problem-solving.

## What is Machine Learning:

Machine Learning is a subset of AI that enables systems to learn from data and improve their performance on tasks without being explicitly programmed.



# Components of Data Science

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## Data Collection:

Gathering data from various sources like databases, sensors, web scraping, APIs, or surveys.

## Data Processing and Cleaning:

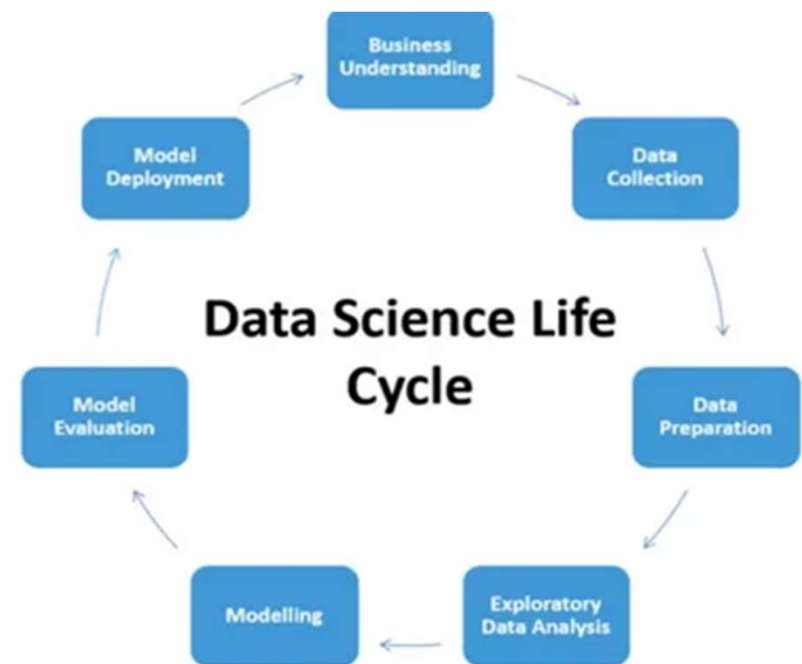
Ensuring data quality by handling missing values, outliers, and inconsistencies.

Techniques include:

- normalization
- standardization
- imputation.

## Exploratory Data analysis:

Using statistical and visualization tools (e.g., histograms, scatter plots) to uncover patterns, correlations, and trends in the data.



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## Modeling:

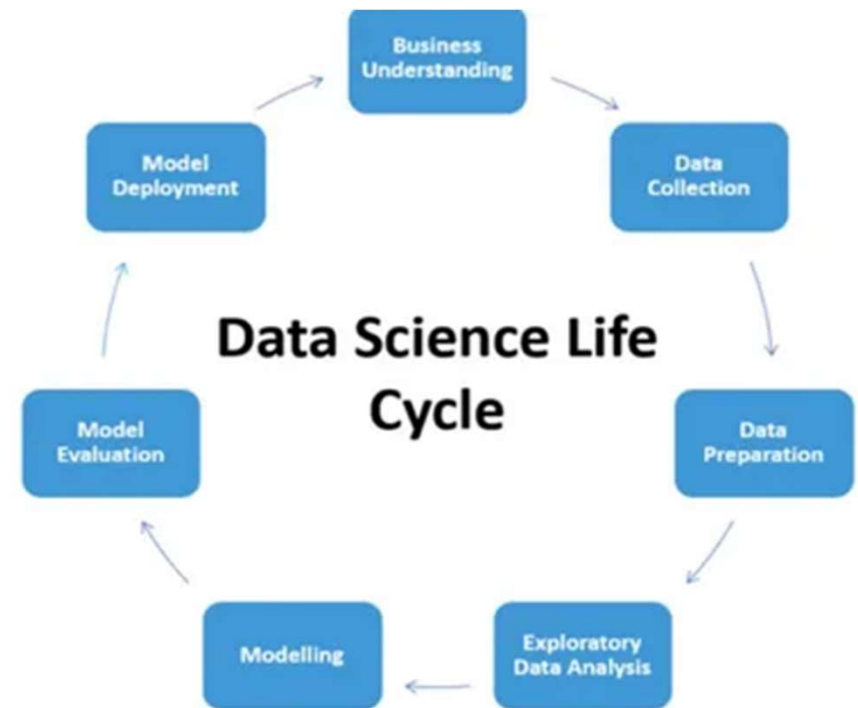
Using algorithms (e.g., regression, decision trees, neural networks) to create models that predict outcomes or classify data.

## Interpretation:

Presenting insights through dashboards, graphs, and charts using tools like Matplotlib, Seaborn, Tableau, or Power BI.

## Deployment:

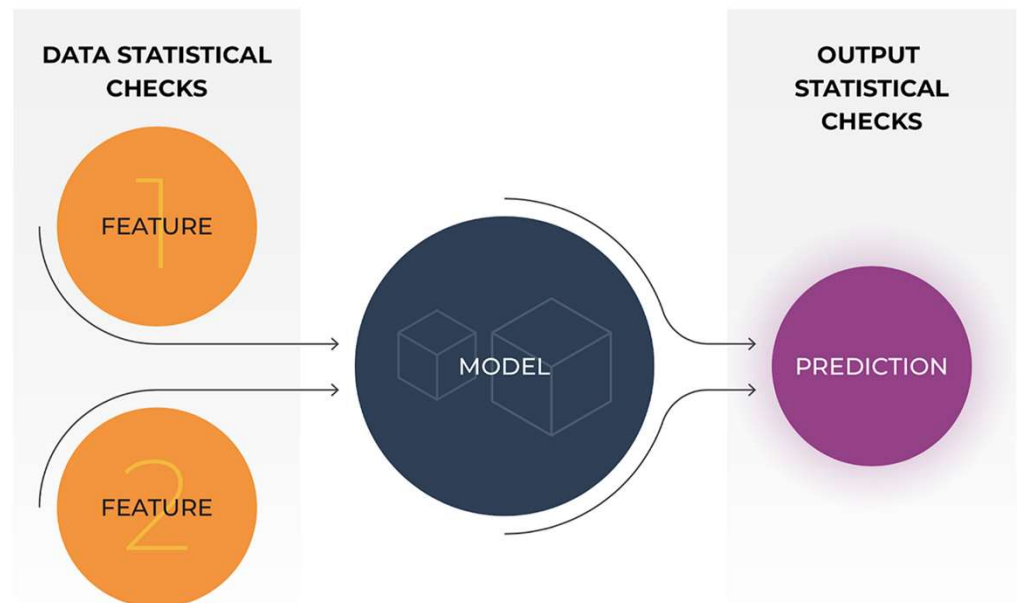
Implementing models in real-world applications (e.g., recommendation systems, fraud detection).



# Machine Learning

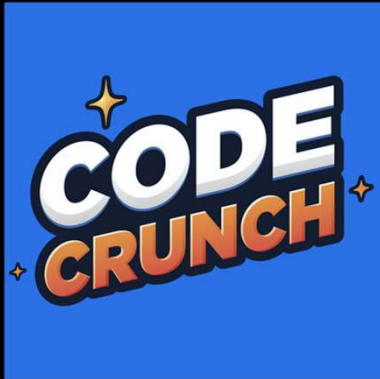
focuses on developing algorithms and models that allow computers to learn from and make predictions or decisions based on data.

- **model** - a mathematical function for describing the relationship between input and outputs for making predictions
- **algorithm** - a procedure or set of decision rules used to carry out a task or prediction





# Q&A Session



# Attendance



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Join us for the next sessions



**Thursdays:** 11AM – 12PM & 2PM – 3PM

**Friday:** 3PM – 4PM

**Monday:** 12PM – 1PM



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